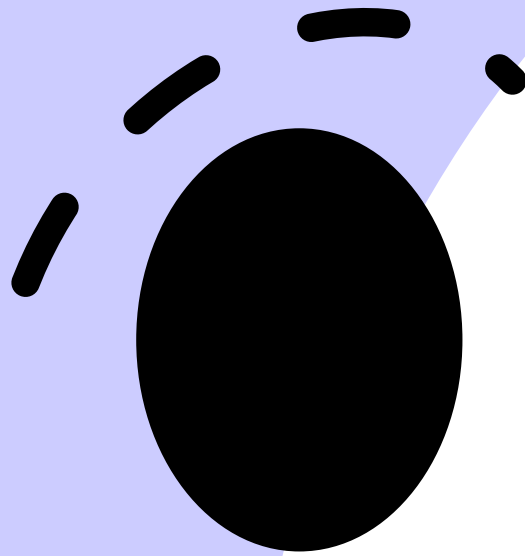
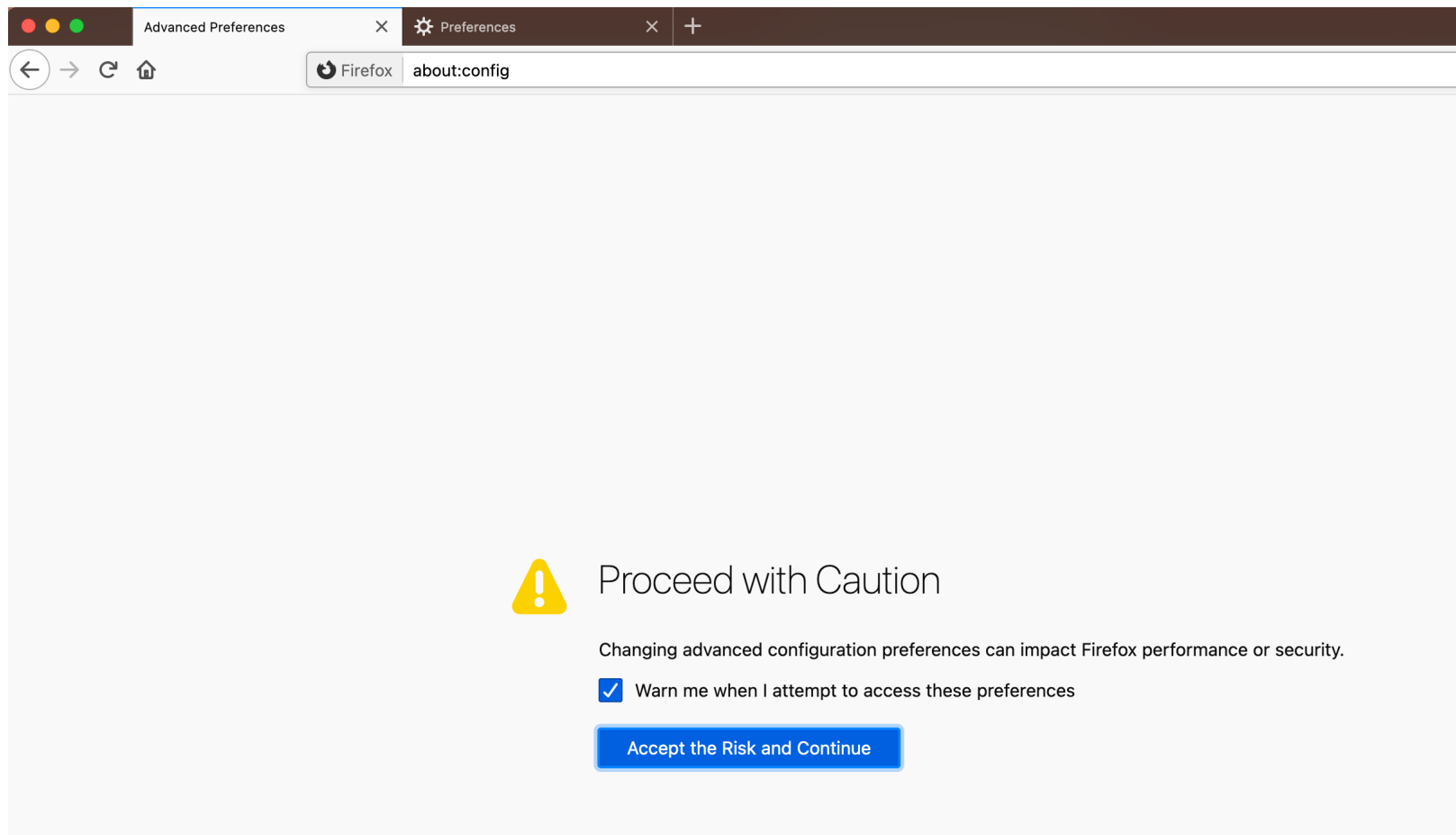




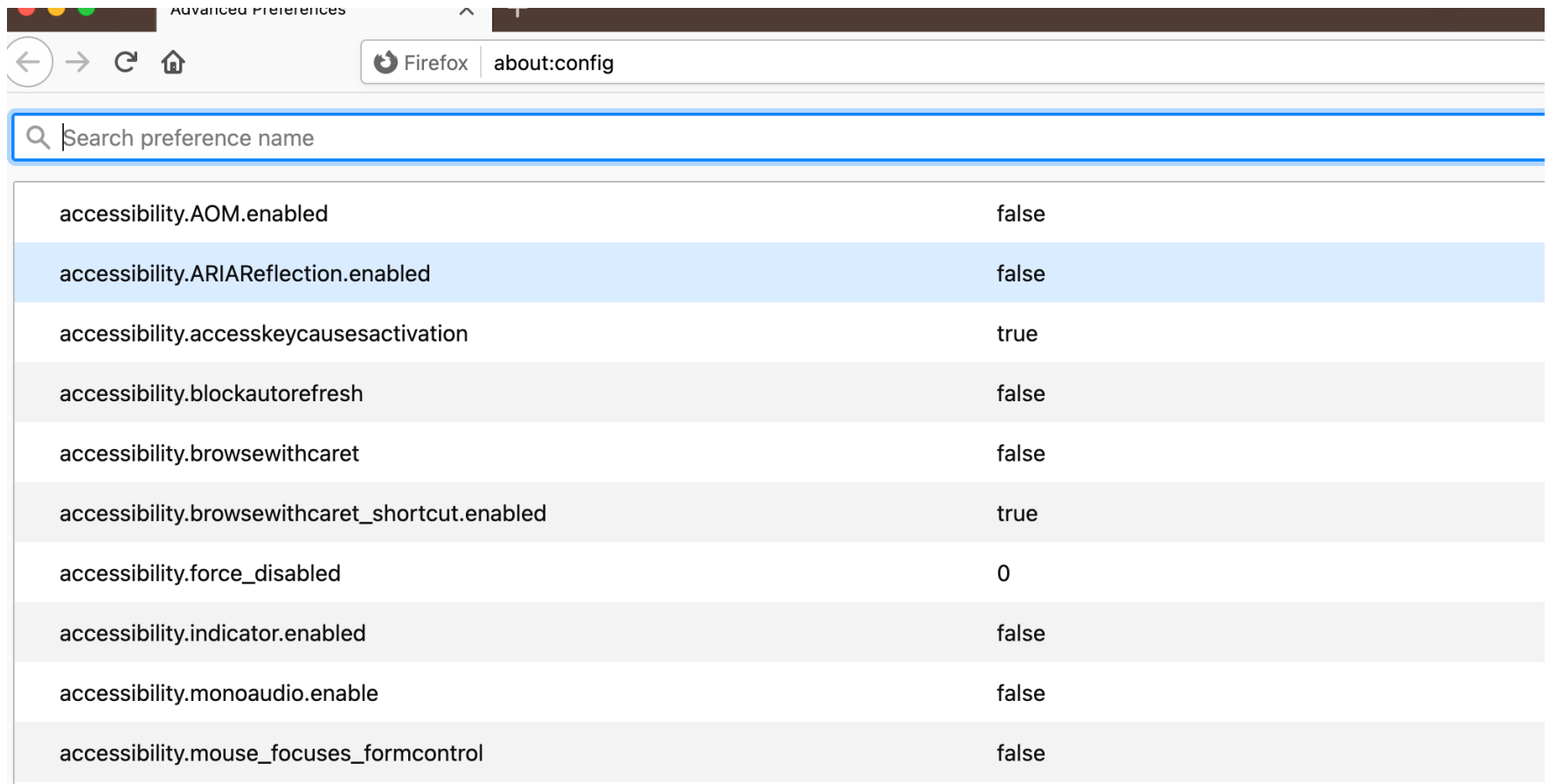
COMS 4170 Software Testing Spring 2026

In Class Exercise

Firefox



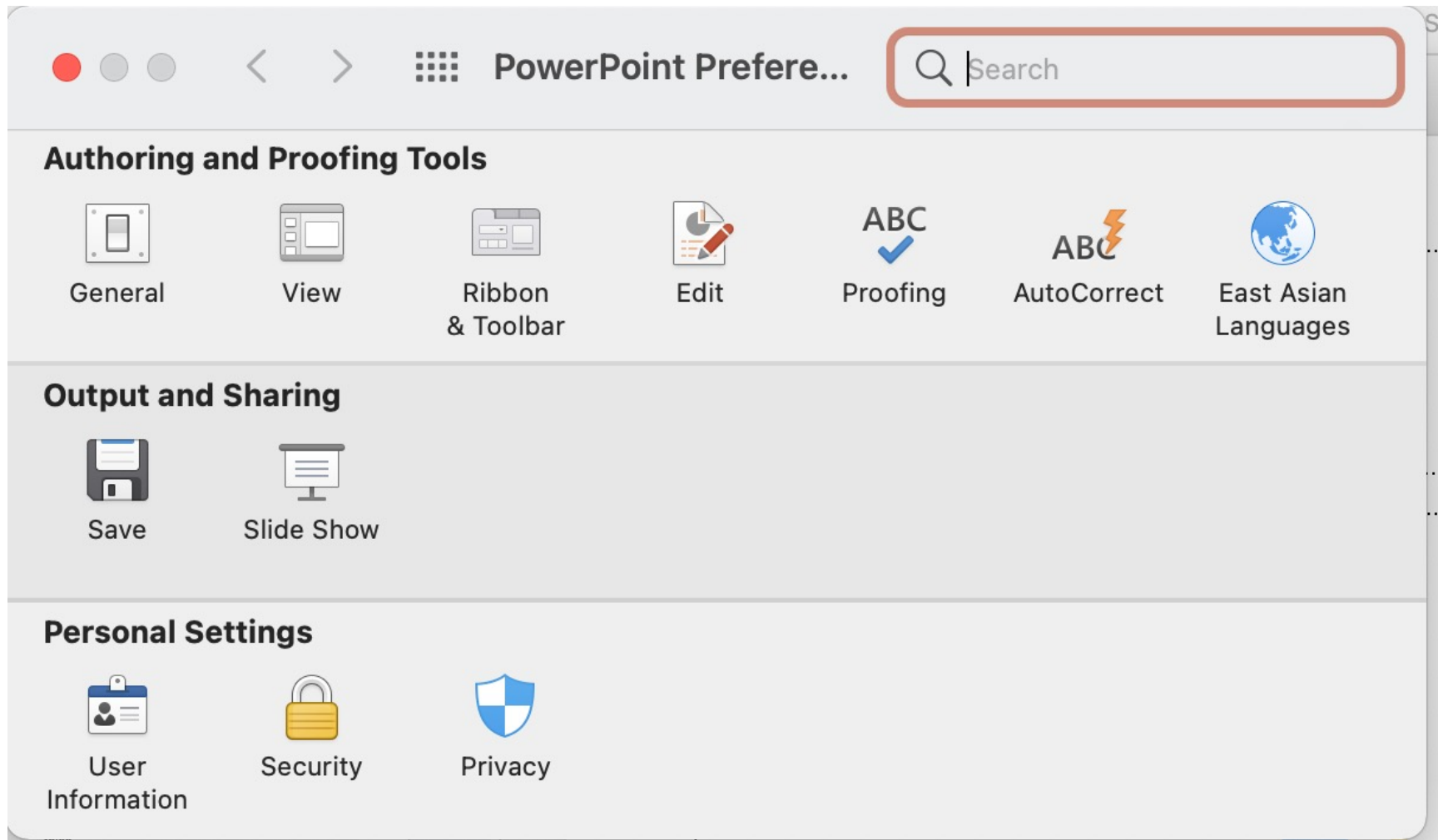
about:config



The image shows a screenshot of the Firefox browser's 'about:config' page. The browser's address bar displays 'Firefox' and 'about:config'. Below the address bar is a search bar with the placeholder text 'Search preference name'. The main content area is a table of configuration preferences. The table has two columns: the preference name and its current value. The rows are alternatingly highlighted with light blue and light gray backgrounds. The preferences listed are all related to accessibility.

Preference Name	Value
accessibility.AOM.enabled	false
accessibility.ARIAReflection.enabled	false
accessibility.accesskeycausesactivation	true
accessibility.blockautorefresh	false
accessibility.browsewithcaret	false
accessibility.browsewithcaret_shortcut.enabled	true
accessibility.force_disabled	0
accessibility.indicator.enabled	false
accessibility.monoaudio.enable	false
accessibility.mouse_focuses_formcontrol	false

PowerPoint



General Tab

Settings

- ☐ Show file extensions
- ☐ Show add-in user interface errors
- ☐ Provide feedback with sound
- ☒ Open Presentation Gallery when opening PowerPoint

Personalize

- ☐ Turn off Dark Mode

Print Quality (Paper/PDF)

Desired Quality

- ☒ High
- ☐ Medium
- ☐ Low

PowerPoint Design Ideas

- ☒ Automatically show me design ideas
Ask for design suggestions any time by clicking Design Ideas on the Design tab.

[Learn more](#)

- ☒ Automatically show me suggestions when I create a new presentation
Suggestions for your deck are shown in the Designer pane when you create a new blank document.

Automatic Alt Text

Automatic alt text generates descriptions for pictures to make them accessible for people with vision impairments. Access alt text at any time by clicking "Edit alt text..." in the context menu for pictures.

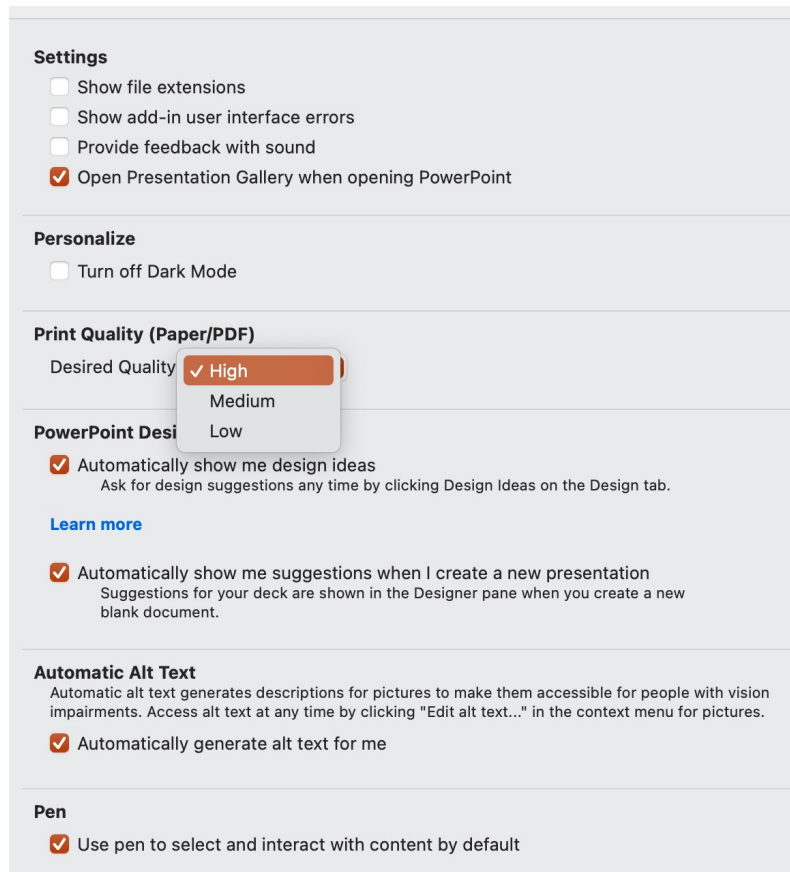
- ☒ Automatically generate alt text for me

Pen

- ☒ Use pen to select and interact with content by default

Lets Model This

Parameters:



The image shows a screenshot of the Microsoft PowerPoint settings window. The window is divided into several sections, each with a title and a list of settings. The 'Settings' section includes options for file extensions, add-in errors, feedback sound, and opening the presentation gallery. The 'Personalize' section has a dark mode toggle. The 'Print Quality (Paper/PDF)' section shows a dropdown menu for 'Desired Quality' with 'High' selected. The 'PowerPoint Designer' section has two checked options for design ideas and suggestions. The 'Automatic Alt Text' section has one checked option for generating alt text. The 'Pen' section has one checked option for using the pen by default.

Settings

- ☐ Show file extensions
- ☐ Show add-in user interface errors
- ☐ Provide feedback with sound
- ☒ Open Presentation Gallery when opening PowerPoint

Personalize

- ☐ Turn off Dark Mode

Print Quality (Paper/PDF)

Desired Quality: High
Medium
Low

PowerPoint Designer

- ☒ Automatically show me design ideas
Ask for design suggestions any time by clicking Design Ideas on the Design tab.
[Learn more](#)
- ☒ Automatically show me suggestions when I create a new presentation
Suggestions for your deck are shown in the Designer pane when you create a new blank document.

Automatic Alt Text

Automatic alt text generates descriptions for pictures to make them accessible for people with vision impairments. Access alt text at any time by clicking "Edit alt text..." in the context menu for pictures.

- ☒ Automatically generate alt text for me

Pen

- ☒ Use pen to select and interact with content by default

A TSL

```
#tsl for powerpoint
```

```
Parameters:
```

```
    Show_Extensions:
```

```
        on.
```

```
        off.
```

```
    Show_add-in:
```

```
        on.
```

```
        off.
```

```
    Sound_feedback:
```

```
        on.
```

```
        off.
```

```
    Open_gallery:
```

```
        on.
```

```
        off.
```

```
    Turn_off_DM:
```

```
        on.
```

```
        off.
```

```
    Print_Quality:
```

```
        high.
```

```
        med.
```

```
        low.
```

```
    Designer:
```

```
        on.
```

```
        off.
```

```
    Suggestions:
```

```
        on.
```

```
        off.
```

```
    Alt_text:
```

```
        on.
```

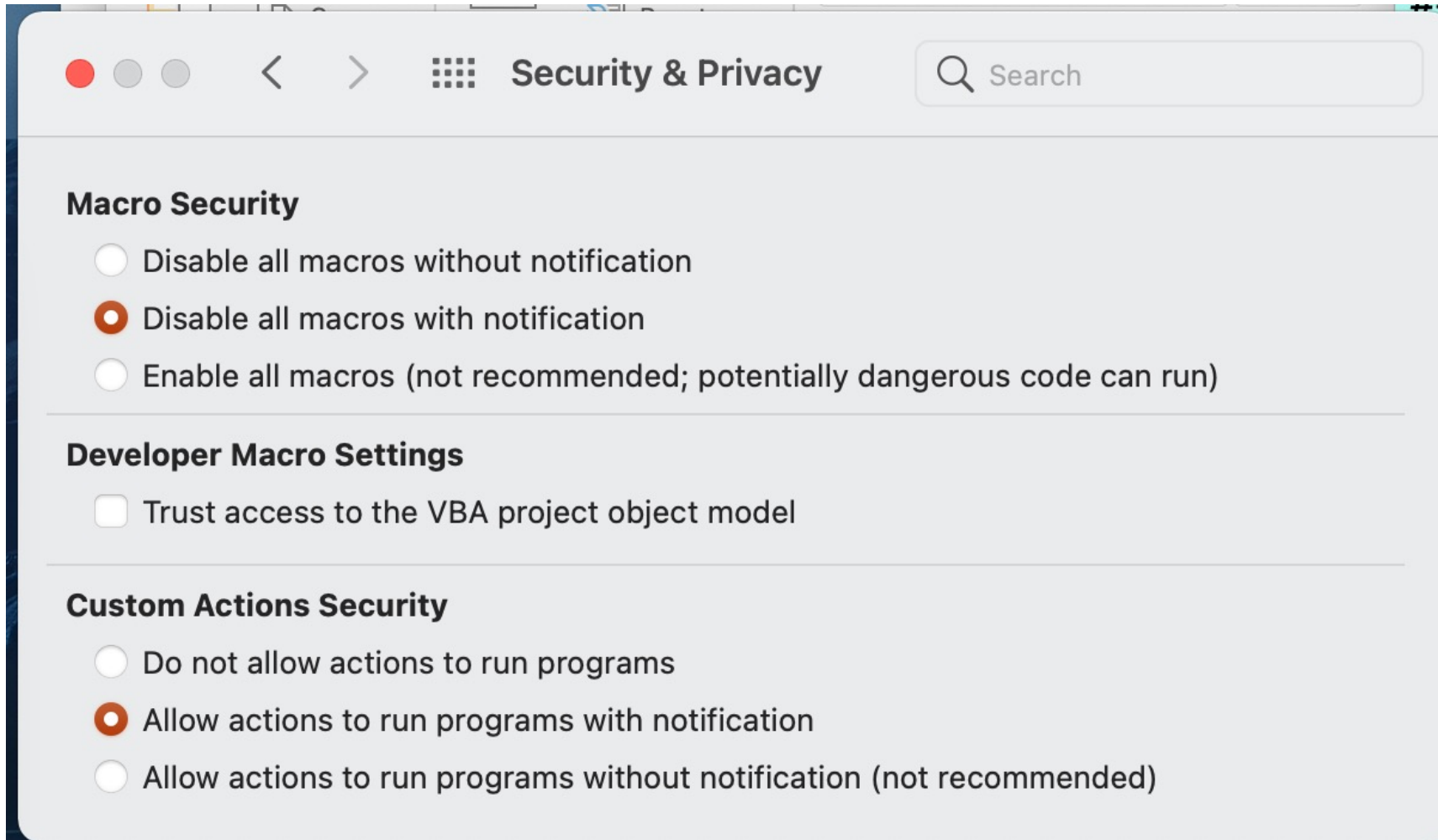
```
        off.
```

```
    Pen:
```

```
        on.
```

```
        off.
```

PowerPoint Security



A TSL

#tsl for powerpoint

Parameters:

Show_Extensions:

on.

off.

Show_add-in:

on.

off.

Sound_feedback:

on.

off.

Open_gallery:

on.

off.

Turn_off_DM:

on.

off.

Print_Quality:

high.

med.

low.

Designer:

on.

off.

Suggestions:

on.

off.

Alt_text:

on.

off.

Pen:

on.

off.

on.

Macro_Security:

disable_without.

disable_with.

enable_all.

Developer_Macro:

on.

off.

Custom_Actions:

do_not_allow.

allow_with.

allow_without.

Running the TSL Tool

```
mcohen>./tsl powerpoint.tsl
```

```
27648 test frames generated and written to powerpoint.tsl.tsl
```

```
mcohen>
```

```
Test Case 1 (Key = 1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.)
```

```
  Show_Extensions : on
  Show_add-in     : on
  Sound_feedback  : on
  Open_gallery    : on
  Turn_off_DM     : on
  Print_Quality   : high
  Designer        : on
  Suggestions     : on
  Alt_text        : on
  Pen             : on
  Macro_Security  : disable_without
  Developer_Macro : on
  Custom_Actions  : do_not_allow
```

```
Test Case 2 (Key = 1.1.1.1.1.1.1.1.1.1.1.1.1.1.2.)
```

```
  Show_Extensions : on
  Show add-in     : on
```

Creating a CIT Sample

```
2
13
2 5
3 1
2 4
3 1
2 1
3 1
```

```
t  k  v  TCount
2 13 29 387
```

```
2 2 2 2 2 3 2 2 2 2 3 2 3
```

```
0  1
2  3
4  5
6  7
8  9
10 11 12
13 14
15 16
17 18
19 20
21 22 23
24 25
26 27 28
```

```
10
0 2 4 7 9 12 13 16 17 19 23 25 27
1 2 5 6 9 11 14 16 17 19 23 24 26
0 2 5 7 9 10 14 16 17 19 22 25 26
0 2 5 6 8 11 14 15 18 20 22 25 27
1 3 4 6 9 12 13 16 18 20 22 24 28
1 3 5 7 8 12 14 15 18 19 21 25 26
0 3 4 7 9 11 13 15 17 20 21 24 26
1 3 4 7 8 10 14 15 18 20 23 25 28
1 3 5 6 8 10 13 16 18 19 21 24 27
0 2 5 7 8 11 13 16 17 19 21 24 28
```

Mapping

```
#tsl for powerpoint
Parameters:
  Show_Extensions:
    on. 0
    off. 1
  Show_add-in:
    on.
    off.
  Sound_feedback:
    on.
    off.
  Open_gallery:
    on.
    off.
  Turn_off_DM:
    on.
    off.
  Print_Quality:
    high.
    med.
    low.
  Designer:
    on.
    off.
  Suggestions:
    on.
    off.
  Alt_text:
    on.
    off.
  Pen:
    on.
    off.
```

t	k	v	TCount
---	---	---	--------

2	13	29	387
---	----	----	-----

2	2	2	2	2	3	2	2	2	2	3	2	3
---	---	---	---	---	---	---	---	---	---	---	---	---

0	1
---	---

2	3
---	---

4	5
---	---

6	7
---	---

8	9
---	---

10	11	12
----	----	----

13	14
----	----

15	16
----	----

17	18
----	----

19	20
----	----

21	22	23
----	----	----

24	25
----	----

26	27	28
----	----	----

10

on

0	2	4	7	9	12	13	16	17	19	23	25	27
---	---	---	---	---	----	----	----	----	----	----	----	----

1	2	5	6	9	11	14	16	17	19	23	24	26
---	---	---	---	---	----	----	----	----	----	----	----	----

0	2	5	7	9	10	14	16	17	19	22	25	26
---	---	---	---	---	----	----	----	----	----	----	----	----

0	2	5	6	8	11	14	15	18	20	22	25	27
---	---	---	---	---	----	----	----	----	----	----	----	----

1	3	4	6	9	12	13	16	18	20	22	24	28
---	---	---	---	---	----	----	----	----	----	----	----	----

1	3	5	7	8	12	14	15	18	19	21	25	26
---	---	---	---	---	----	----	----	----	----	----	----	----

0	3	4	7	9	11	13	15	17	20	21	24	26
---	---	---	---	---	----	----	----	----	----	----	----	----

1	3	4	7	8	10	14	15	18	20	23	25	28
---	---	---	---	---	----	----	----	----	----	----	----	----

1	3	5	6	8	10	13	16	18	19	21	24	27
---	---	---	---	---	----	----	----	----	----	----	----	----

0	2	5	<u>7</u>	8	11	13	16	17	19	21	24	28
---	---	---	----------	---	----	----	----	----	----	----	----	----

Base Choice simplified PowerPoint

Parameters:

Open_gallery:

on.

off.

Turn_off_DM:

on.

off.

Print_Quality:

high.

med.

Pen:

on.

off.

Custom_Actions:

do_not_allow.

allow_with.

allow_without.

Base:

OG: -on

Toff: -on

Print: med

Pen: off

Custom: don't allow

$$N=1+(1+1+1+1+2)=7$$

Parameters:

- Open_gallery:
 - on.
 - off.
- Turn_off_DM:
 - on.
 - off.
- Print_Quality:
 - high.
 - med.
- Pen:
 - on.
 - off.
- Custom_Actions:
 - do_not_allow.
 - allow_with.
 - allow_without.

OG	Toff	Print	Pen	custom
On	On	Med	Off	Don't allow
On	Off	Med	Off	Don't allow
On	On	High	Off	Don't' allow
On	On	Med	On	Don't allow
On	On	Med	Off	Allow-with
On	On	Med	Off	Allow- without
Off	On	Med	Off	Don't Allow